Free balancing equations worksheet worksheet 19

Maths | 7th grade | Algebra

1. If 4x + 3 = 19, what is the value of x?

A) 2
B) 3
C) 4
D) 5

2. The expression 3(y + 2) equals 18. What is the value of y?

A) 2
B) 3
C) 4
D) 5

3. Solve for z: 5z - 2 = 3z + 10.

A) 5
B) 6
C) 7
D) 8

4. What is the value of x in the equation 7x + 4 = 3x + 20?

A) 3
B) 4
C) 5
D) 6

5. Which value of x makes the equation 2(x - 3) = x + 4 true?

A) 7
B) 8
C) 9
D) 10

6. Find the value of y if 2(y - 1) + 3 = 11.

A) 4
B) 5
C) 6
D) 7

7. What is the solution to the equation 6a + 9 = 4a + 15?

A) 2
B) 3
C) 4
D) 5

8. Which of the following values for b satisfies the equation b/2 - 5 = 2?

A) 10
B) 12

C) 14
D) 16

9. Solve 3(m + 2) = 2m + 9 for m.

A) 1
B) 2
C) 3
D) 4

10. Determine the value of x in the equation 8 - 2(x + 1) = 4.

A) 0
B) 1
C) 2
D) 3

Answer Key

1. C
Subtract 3 from both sides to get 4x = 16, then divide by 4 to find x = 4.

2. C
Distribute to get 3y + 6 = 18. Subtract 6 to get 3y = 12, then divide by 3 to find y = 4.

3. A
Subtract 3z from both sides to get 2z - 2 = 10. Add 2 to both sides to get 2z = 12, then divide by 2 to find z = 5.

4. D
Subtract 3x from both sides to get 4x + 4 = 20. Subtract 4 to get 4x = 16, then divide by 4 to find x = 6.

5. A
Distribute to get 2x - 6 = x + 4. Subtract x from both sides to get x - 6 = 4, then add 6 to find x = 10.

6. B
Simplify to get 2y - 2 + 3 = 11. Combine like terms to get 2y + 1 = 11. Subtract 1 to get 2y = 10, then divide by 2 to find y = 5.

7. A
Subtract 4a from both sides to get 2a + 9 = 15. Subtract 9 to get 2a = 6, then divide by 2 to find a = 3.

8. C
Add 5 to both sides to get b/2 = 7, then multiply both sides by 2 to find b = 14.

9. B
Distribute to get 3m + 6 = 2m + 9. Subtract 2m from both sides to get m + 6 = 9, then subtract 6 to find m = 3.

10. D
Distribute to get 8 - 2x - 2 = 4. Simplify to 6 - 2x = 4. Subtract 6 to get -2x = -2, then divide by -2 to find x = 1.